

Department of Computer and Software Engineering**SE: Machine Learning****Course Instructor:****Date:****Day:****Semester:****Lab 10: Model Optimization and Feature Engineering**

Lab Title	CLO	BT	PLO	Weightage
Model Optimization & Feature Engineering CV, Optuna, Encoding & Leakage				

Name	Roll Number	Report /10	Viva /5	Total /15

Checked on: _____

Signature: _____

Objectives

- Apply Grid Search, Random Search, and Bayesian Optimization
- Understand and use Cross-Validation properly
- Implement Target Encoding and avoid leakage
- Identify and fix Data Leakage
- Engineer Cyclical Features

Problem Statement

You are given a dataset for a binary classification problem. The dataset contains numerical, categorical, and temporal features.

Your goal is to improve model performance using optimization techniques and feature engineering while ensuring that no data leakage occurs.

Dataset File: `lab10_data.csv`

The dataset contains:

- Numerical features: age, income
- Categorical feature: city
- Temporal feature: hour (0–23)
- Target variable: binary classification
- A hidden leakage feature for analysis

Dataset: Customer Transaction Response Prediction

This dataset simulates a real-world customer behavior prediction problem commonly encountered in domains like digital marketing, fraud detection, recommendation systems, and customer analytics.

Each row represents a user interaction or transaction event, and the goal is to predict a binary outcome (e.g., whether a user converts, responds, or triggers a flag).

Feature Description

Numerical Features

Numerical features include **age**, representing the age of the customer (18–60), which often correlates with purchasing power, risk behavior, or engagement; and **income**, a simulated

annual income with realistic variance where higher-income users tend to behave differently in spending, fraud patterns, or response likelihood.

Categorical Feature

The categorical feature **city** represents customer location segment (A, B, C) and in practice could correspond to geographic regions, customer tiers, or market segments. This feature is useful for demonstrating target encoding, categorical bias, and leakage pitfalls.

Temporal Feature (Cyclical)

The temporal feature **hour (0–23)** represents the hour of the event (e.g., transaction time); in real-world systems user behavior is strongly time-dependent, fraud often spikes at unusual hours, and engagement varies across the day. This feature is intentionally included to teach why raw hour is a poor representation and why cyclical encoding (sin/cos) is necessary.

Target Variable

The **target (0 or 1)** represents a binary outcome such as fraud vs normal transaction, click vs no click, or conversion vs no conversion. The target is not purely random—it depends on income patterns and time-based behavior (hour), ensuring models can learn meaningful patterns and that optimization techniques show measurable improvements.

Part A: Hyperparameter Optimization

Task 1: Grid Search

Use GridSearchCV to tune a Decision Tree model.

Tune: - max_depth - min_samples_split

Report the best parameters.

Task 2: Random Search

Use RandomizedSearchCV.

Compare results and runtime with Grid Search.

Task 3: Bayesian Optimization (Optuna)

Optimize hyperparameters using Optuna.

Run at least 20 trials and report best parameters.

Part B: Cross-Validation

Task 4: K-Fold Cross Validation

Perform 5-fold cross-validation.

Report: - Mean accuracy - Standard deviation

Task 5: Comparison

Compare: - Train/Test split - Cross-validation

Discuss differences in performance.

Part C: Target Encoding

Task 6: Naive Target Encoding

Encode the categorical feature using mean target value.

Observe model performance.

Task 7: K-Fold Target Encoding

Implement K-Fold target encoding to avoid leakage.

Compare with naive encoding.

Part D: Data Leakage

Task 8: Identify Leakage

Identify features or steps that introduce leakage.
Explain why they are problematic.

Task 9: Fix Leakage

Modify the pipeline to remove leakage.
Compare performance before and after.

Part E: Cyclical Features

Task 10: Cyclical Encoding

Transform the hour feature using:

$$\sin\left(\frac{2\pi x}{24}\right), \quad \cos\left(\frac{2\pi x}{24}\right)$$

Task 11: Comparison

Compare model performance: - Before encoding - After encoding

Discussion Questions

- Why does cross-validation give more reliable estimates than a single split?
- How does target encoding cause leakage?
- Why are cyclical features better represented using sine/cosine?
- Which optimization method performed best and why?